Robert Leggieri

APPLIED AI | MACHINE LEARNING | TECHNOLOGY LEADERSHIP

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PROFESSIONAL SUMMARY

Technology leader and applied AI practitioner with over two decades of systems ownership and graduate training in artificial intelligence. Builds secure, auditable systems across natural language processing, speech, computer vision, and data engineering. Combines hands-on model implementation with operational judgment, security awareness, and cross-functional delivery.

TECHNICAL SKILLS

AI / ML: PyTorch, TensorFlow / Keras, scikit-learn, Hugging Face Transformers, CNNs, multimodal learning, model evaluation, LoRA / QLoRA

Data / Engineering: Python, NumPy, Pandas, OpenCV, SQL / SQLite, REST APIs, OAuth 2.0 / PKCE, ETL pipelines, pytest, Git

Systems / Automation: PowerShell, Windows Server, network and security operations, workflow automation, vendor and infrastructure strategy

SELECTED AI PROJECTS

Text-Prosody Alignment: The "I'm Fine" Effect

Case Studies in Machine Learning, The University of Texas at Austin

- Created the Alignment Discrepancy Score (ADS) to quantify text-sentiment and audio-prosody mismatch in DAIC-WOZ interviews; observed approximately 0.62 AUC for elevated PHQ-8 classification above audio-only baselines.
- Published text-prosody-evidence, a separate dataset-neutral reference implementation for leakage-aware evaluation, calibration, reliability, bootstrap intervals, provenance, and auditable evidence manifests.

BQE CORE AI Data Connector

Professional project, Gutschick, Little & Weber, P.A. | GitHub

- Built a governed read-only ingestion layer that transforms BQE CORE operational and financial data into normalized, auditable structures for downstream AI-assisted analysis.
- Implemented OAuth / PKCE, secure credential storage, pagination, retries, deduplication, incremental sync, SQLite analytics, reconciliation, and raw-response audit preservation.

PraxiCom: Affect-Aware Simulated Patient

UT Austin NURSING-AI Challenge | GitHub | Demo

- Developed a voice-based simulated patient with nursing-faculty guidance using React, Node.js, Socket.IO, YAML personas, Hume EVI, live transcripts, faculty chat, and structured evaluation; used synthetic patients and no PHI.

Advances in Deep Learning - Four-Project Series

The University of Texas at Austin

- Implemented mixed precision, LoRA, 4-bit quantization, and QLoRA; built patch autoencoding, Binary Spherical Quantization, and autoregressive models for generative image compression.
- Adapted smaller LLMs for mathematical reasoning and generated nearly one million grounded vision-language QA pairs from SuperTuxKart game state.

Debiasing an NLI Model

AI 388 Natural Language Processing Final Project, The University of Texas at Austin

- Used dataset cartography and loss re-weighting to reduce spurious shortcuts in MNLI, improving generalization on adversarial and out-of-domain examples.

PROFESSIONAL EXPERIENCE

Network Manager - Gutschick, Little & Weber, P.A.

Burtonsville, MD | July 2004 - Present

- Own technology operations, network infrastructure, security, automation, vendor strategy, and long-range planning for an engineering firm supporting more than 40 users.
- Use Python and PowerShell to automate workflows and integrate systems; translate business requirements into reliable, secure, and cost-conscious technology solutions.

EDUCATION

The University of Texas at Austin - M.S., Artificial Intelligence, GPA: 4.0 | Expected August 2026

University of Maryland Global Campus - B.S., Data Science, Summa Cum Laude | May 2024